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Studierichting: Informatica
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$$I = \int x^9 \sqrt{x^5+1} dx$$

Stel $x^5 + 1 = t^2$

$$\Rightarrow 5x^4 dx = \frac{2}{5} t dt$$

$$I = \int x^4 x^5 \sqrt{x^5+1} dx$$

$$= \frac{2}{5} \int \left(t^2 - 1\right) t^2 dt$$

$$= \frac{2}{5} \left(\frac{t^5}{5} - \frac{t^3}{3}\right) + C$$

$$= \frac{2\left(x^5+1\right)^{\frac{5}{2}}}{25} - \frac{2\left(x^5+1\right)^{\frac{3}{2}}}{15} + C$$

References